SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA13**

COURSE OUTCOME COVERED

CO1: ***Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems(BL:3)***

Assessment Tool Weightage: 10%

QUESTIONS: 4 **Total Marks:20**
Duration : 1 Hour

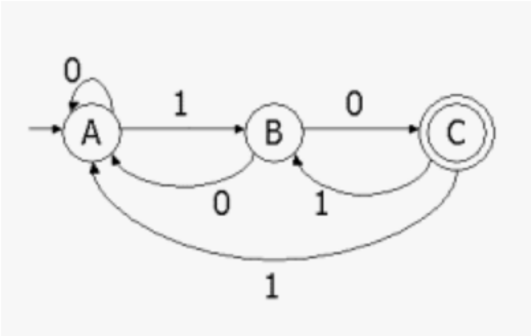
Technical Presentation

Code of Conduct:

I _____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

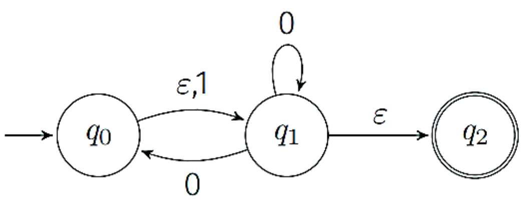
1.How can every Non-Deterministic Finite Automaton (NFA) be converted into an equivalent Deterministic Finite Automaton (DFA), and what are the challenges involved? Consider the NFA given below which accepts some language L over the alphabet Σ={0,1}. Construct a DFA that represents the same language L



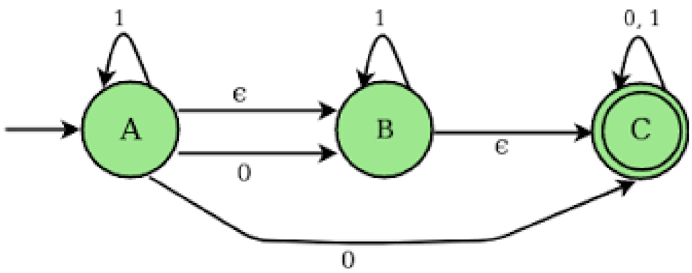
2.How is mathematical induction used to prove properties of strings and languages in automata theory? Apply the principle of mathematical induction and prove that for all positive integers n, the statement given below is true.

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

3.What role do epsilon (ε) transitions play in finite automata, and how can ε-NFA be converted into an equivalent NFA without epsilon transitions? construct an equivalent NFA without ε-transitions that accepts the same language. Draw its transition table and mark the initial and final states



4.A NFA with ε-transitions is given below. Find ε-closure for each state and construct an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (8 Marks)	Level 3 – Proficient (6 Marks)	Level 2 – Developing (4 Marks)	Level 1 – Beginning (2 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandabl e with minor issues	Somewhat unclear or poorly structured	Difficult to understand